Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Product-focused Software Engineer with 2+ years shipping **scalable, real-time applications**, customer-facing features, and production services using Python, TypeScript, React, Node.js, PostgreSQL, Redis, Kafka, WebSockets, cloud infrastructure, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**

Master of Science in Computer Science *GPA: 3.75/4*

Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**

Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**

Software Engineer *Los Angeles, CA*

- Owned a **multi-tenant product platform** from data modeling through React dashboards, reducing latency from 1.2 seconds to 350 milliseconds while supporting 500+ daily users and evolving requirements.
- Reworked asynchronous services with explicit job states, failure isolation, and **production monitoring**, tripling throughput for 1K+ concurrent jobs while maintaining 99%+ reliability across 5K+ monthly production runs.

Grad8 **June 2025 – August 2025**

Software Engineer Intern *Los Angeles, CA*

- Shipped **React and Flask** features for uploads, grading, history, and feedback, supporting 150+ instructors and 10K+ daily requests through responsive interfaces and tested production APIs reliably at scale.
- Implemented concurrent processing and automatic model fallbacks for 1K+ files per batch, reducing **user turnaround time** 60% while deploying changes through Git reviews and CI/CD safely in production.

Aasma Technology Solutions **August 2023 – July 2024**

Software Developer Engineer – SecureSync *India*

- Simplified distributed integration services through **object-oriented architecture**, reducing API latency 30%, technical debt 20%, and duplicated logic while improving maintainability across a reliable, scalable cross-platform data-sharing product.
- Rebuilt AWS Glue and S3 **event-processing pipelines**, improving efficiency 68%, then maintained 20+ automated tests and Jenkins releases across diverse Windows, Linux, and macOS production environments worldwide.

Aasma Technology Solutions **May 2023 – August 2023**

Full Stack Developer Intern – VibrantChem *India*

- Built a **React and Node.js** operations product with REST APIs, WebSockets, permissions, and notifications, synchronizing 2M+ records across locations while reducing staged-release conflicts 33% during major production launches.
- Owned eight services from implementation through **Docker and Jenkins** launch, adding automated validation and monitoring that reduced release time 50% while sustaining 95%+ reliability for customer workflows.

LST Capitals **November 2022 – January 2023**

Web Developer Intern *India*

- Built **transactional booking and payment flows** with React, Flask, SQL, coupons, and real-time status updates, reliably serving 50K+ records through user-facing APIs with sub-200-millisecond responses at scale.
- Designed 12+ **third-party integrations** for payments, synchronization, and webhooks, presenting prototypes and incorporating client feedback before implementation to consistently reduce design errors 25% across product delivery cycles.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, HTML5, CSS3
Product & Frontend: React, Next.js, Responsive Design, WebSockets, Real-Time UX, Growth Experiments, A/B Testing, Accessibility
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, Microservices, Async Processing, Event-Driven Systems
Scale & Data: PostgreSQL, Redis, Kafka, RabbitMQ, AWS, Kubernetes, Docker, WebRTC, Load Testing, Rate Limiting
Engineering: Git, Linux, CI/CD, GitHub Actions, Jenkins, PyTest, Jest, Monitoring, Debugging, Incident Triage, Code Reviews, Agile

Projects

LiveBid Marketplace | *Python, React, WebSockets, Kafka, Redis, PostgreSQL, Stripe*

- Built a **real-time auction marketplace** with synchronized bids, idempotent checkout, seller inventory, and live notifications, using Kafka ordering and Redis state to preserve correctness during 10K-user traffic simulations.

StreamScale Control Plane | *Python, WebRTC, Kubernetes, Redis, Prometheus, Grafana*

- Built a **livestream reliability platform** with admission control, connection pooling, regional failover, and progressive load testing, sustaining 100K simulated viewers while maintaining sub-second startup and actionable telemetry.

FinNexus | *Next.js, TypeScript, MongoDB, Plaid, WebSockets, Kubernetes*

- Built a **real-time financial product** integrating Plaid accounts, transaction synchronization, budgeting dashboards, and notifications, using TypeScript services and Kubernetes deployment to deliver reliable, secure multi-account user workflows.